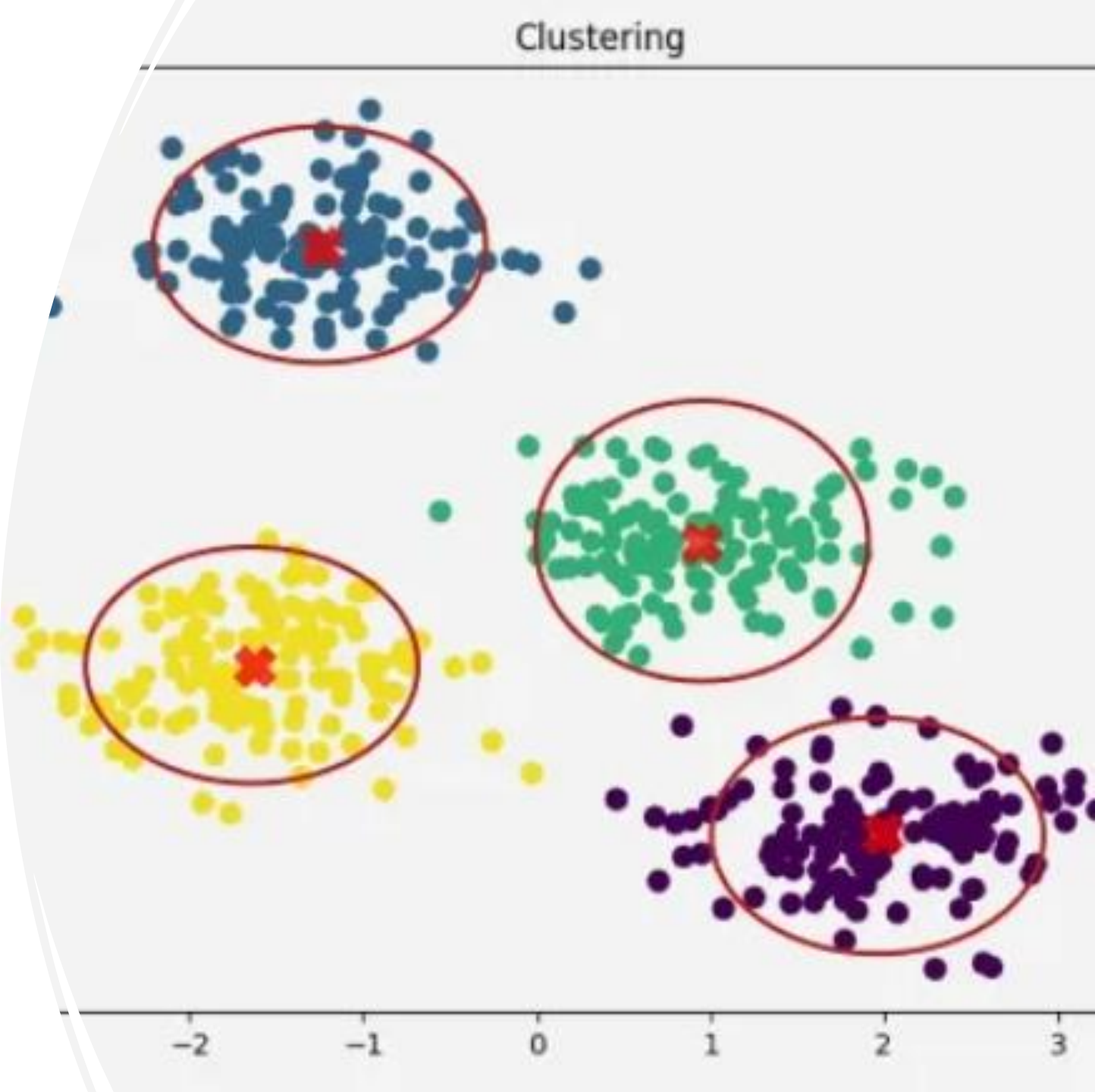
Unsupervised Learning (Clustering)

- Clustering is an unsupervised machine learning method that organizes data points into groups (clusters) based on their similarities, without relying on labeled examples. The aim is to group together data points that are highly similar while keeping dissimilar points in separate clusters, helping uncover hidden patterns and structures within complex datasets.
- **Goal:** Identify natural groupings or underlying structures in unlabeled data without predefined classes.
- **How it works:** Data points are grouped into clusters using similarity or distance-based criteria.
- **Similarity measures:** Common metrics include Euclidean distance, cosine similarity, and others, depending on the data and clustering approach.
- **Output:** Each data point is assigned a cluster ID that represents a group of shared characteristics.

Applications

- For example, with customer purchase data, clustering can be used to group customers who exhibit similar buying behaviors. These groups can then support targeted marketing campaigns, personalized product recommendations, and effective customer segmentation strategies.



K-Means Clustering

- -Means clustering is an unsupervised learning technique that organizes data points into groups without requiring labeled data. It is commonly used to reveal hidden patterns by grouping data based on similarity.
- Identifies natural groupings in unlabeled datasets
- Groups data points according to their distance from cluster centroids
- Widely applied in customer segmentation, image compression, and pattern recognition
- Effective for extracting structure from raw, unorganized data

Working of K-Means Clustering

Consider a dataset consisting of items described by a set of features, where each item can be represented as a feature vector. The objective is to group these items into meaningful categories. To do this, we use the K-Means clustering algorithm.

In K-Means, **K** represents the number of clusters or groups into which the data is to be divided. The algorithm works by organizing the data points into K clusters based on their similarity, ensuring that items within the same cluster are more similar to each other than to those in other clusters.

Working of K-Means Clustering(STEPS)

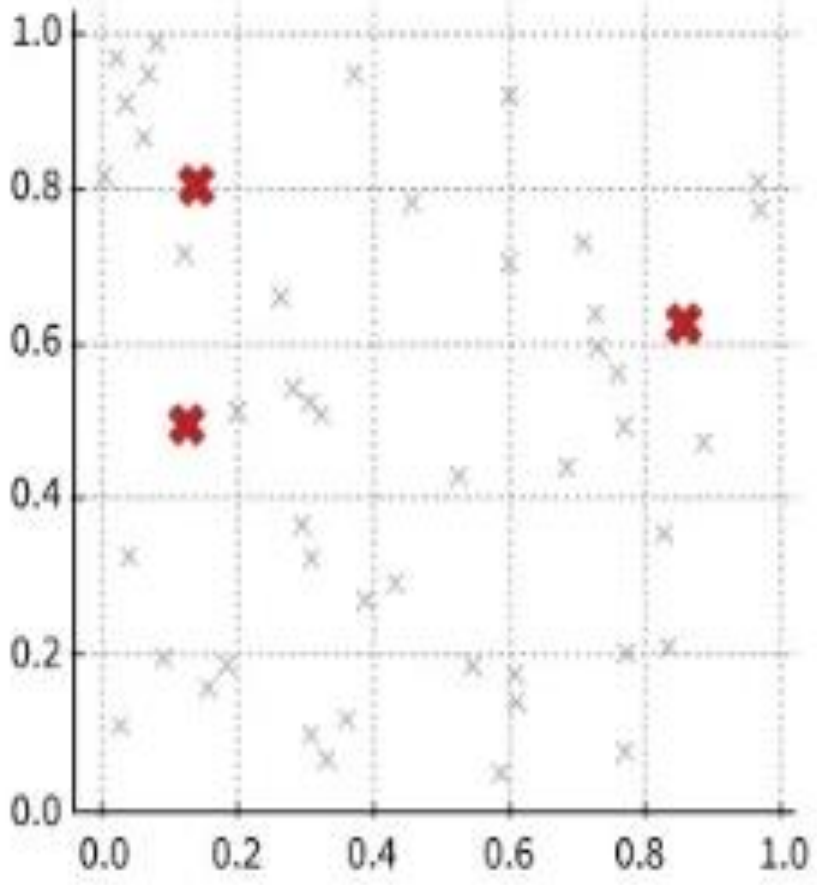
- The algorithm divides the items into **K** groups or clusters based on similarity. To measure this similarity, Euclidean distance is commonly used. The K-Means algorithm follows these steps:
- **Initialization:** Randomly select **K** initial cluster centroids.
- **Assignment step:** Assign each data point to the nearest centroid, forming **K** clusters.
- **Update step:** Recalculate each centroid by computing the average of all data points assigned to that cluster.
- **Repeat:** Continue the assignment and update steps until the centroids stabilize or a maximum number of iterations is reached.
- The objective is to partition the dataset into **K** clusters such that data points within the same cluster are more similar to each other than to those in different clusters.

STEP 1

Choose Initial Centroids

Centroids are randomly chosen from the data points. These represent the initial cluster centers.

✖ Centroids × Data Points

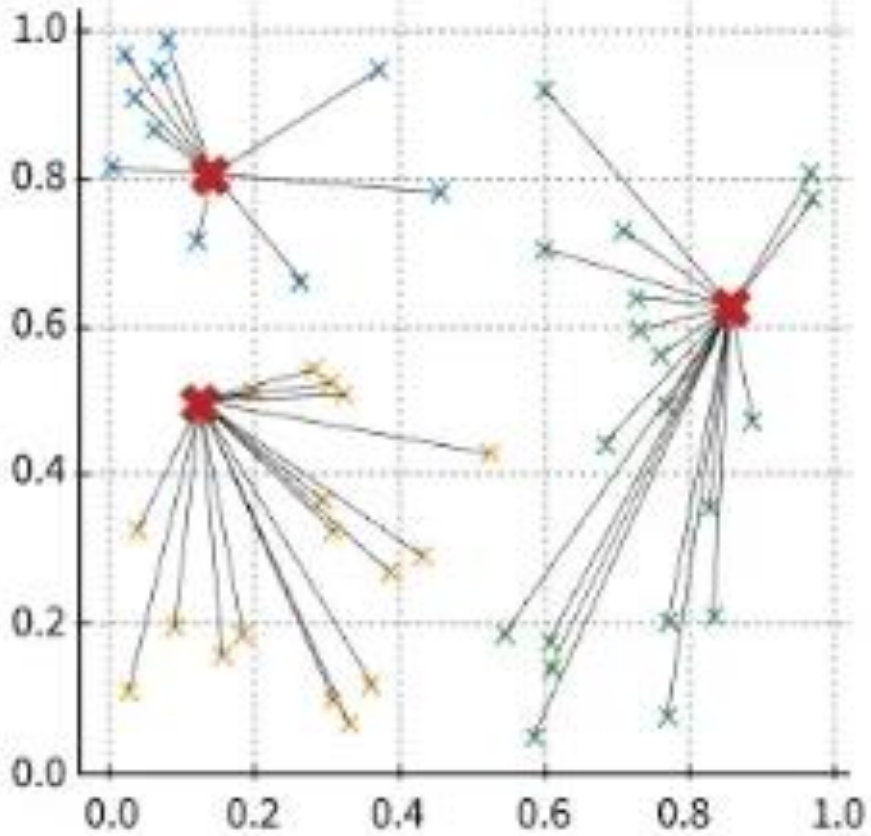


STEP 2

Assign Points to Nearest Centroid

Each point is assigned to the nearest centroid, forming clusters

- Centroids
- Cluster 1
- Cluster 2
- Cluster 3

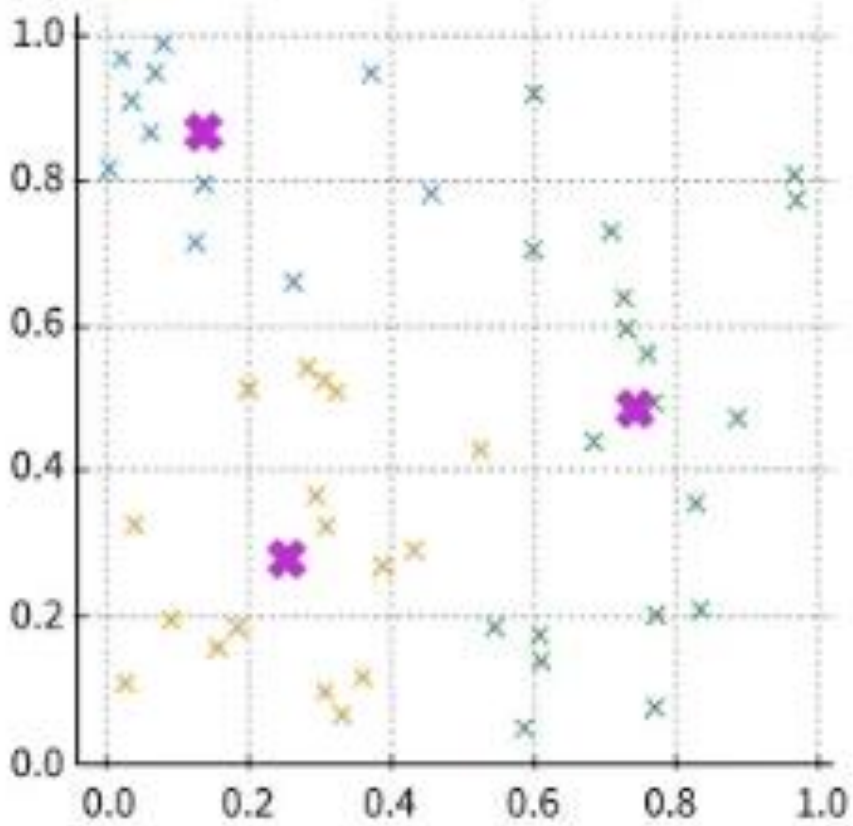


STEP 3

Update Centroids

Centroids are recalculated as the mean of the points in each cluster

- New Centroids
- Cluster 1
- Cluster 2
- Cluster 3

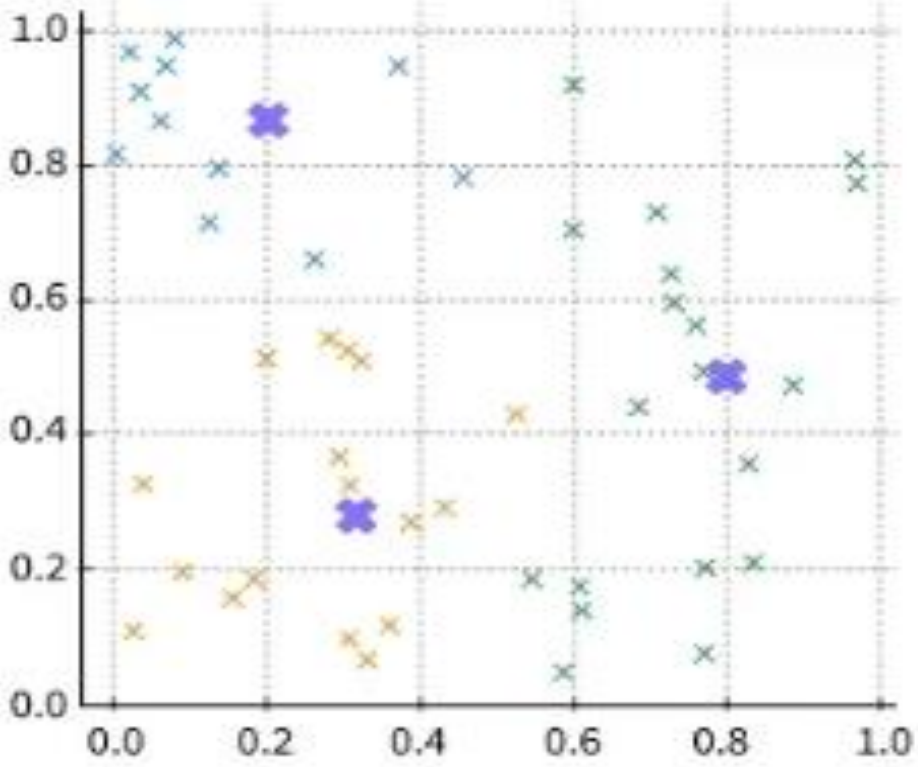


STEP 4

Repeat Until Convergence

This process repeats until the centroids stabilize and do not move further.

- Final Centroids
- Cluster 1
- Cluster 2
- Cluster 3



Why Use K-Means Clustering?

- K-Means is widely used across many applications because of its simplicity, efficiency, and reliable performance. Some key reasons for its popularity include:
- **Data Segmentation:** K-Means is commonly used to divide data into meaningful groups. For instance, businesses apply it to segment customers based on behavior such as purchasing habits or website interactions.
- **Image Compression:** The algorithm can reduce image complexity by clustering similar pixels together, resulting in effective image compression for storage and processing.
- **Anomaly Detection:** K-Means helps identify anomalies or outliers by detecting data points that do not fit well into any cluster.
- **Document Clustering:** In natural language processing (NLP), K-Means is used to group similar documents or articles, supporting applications like recommendation systems and news categorization.
- **Organizing Large Datasets:** For large datasets, K-Means helps structure data into smaller, more manageable groups based on similarity, making analysis more efficient.

Challenges with K-Means Clustering

- Despite its advantages, the K-Means algorithm has several limitations:
- **Choosing the Right Number of Clusters (K):** Determining the appropriate number of clusters is one of the main challenges and often requires additional methods or experimentation.
- **Sensitivity to Initial Centroids:** The final clustering results can differ depending on the initial random selection of centroids.
- **Assumption of Spherical Clusters:** K-Means assumes clusters are spherical and similar in size, which can be problematic when real-world data contains clusters with varying shapes or densities.
- **Sensitivity to Outliers:** Outliers can significantly affect centroid positions, leading to distorted clusters and less accurate results.

K-Medoids Clustering

- Partitioning Around Medoids (PAM) is a clustering technique similar to K-Means, but it uses **medoids** instead of means to form clusters. A medoid is an actual data point within a cluster that best represents the other points in that cluster, making the method more robust to noise and outliers.
- The steps involved in the K-Medoids (PAM) algorithm are:
- **Initialization:** Randomly select **K** data points as the initial medoids.
- **Assignment:** Assign each data point to the nearest medoid based on a chosen distance measure.
- **Update:** For each cluster, select a new medoid that minimizes the total distance between itself and all other points in the cluster.
- **Repeat:** Continue the assignment and update steps until the medoids no longer change or convergence is achieved.

Advantages and Disadvantages

- **Advantages of K-Medoids**
- **Robustness to Outliers:** Since medoids are actual data points, the method is less sensitive to noise and outliers compared to centroid-based approaches.
- **Flexible Distance Metrics:** K-Medoids can work with various distance measures, not just Euclidean distance, making it suitable for different types of data.
- **Disadvantages of K-Medoids**
- **Computationally Intensive:** K-Medoids is slower than K-Means, especially for large datasets, due to its higher computational complexity.
- **Greater Complexity:** The algorithm is more complex to implement and understand compared to K-Means, despite often producing more reliable clustering results.

Key Differences Between K-Means and K-Medoids

- **entroid vs. Medoid**
- **K-Means:** Uses the average (mean) of all points in a cluster as the centroid, which may not correspond to an actual data point.
- **K-Medoids:** Selects actual data points as medoids, making the clusters easier to interpret.
- **Distance Measures**
- **K-Means:** Primarily relies on Euclidean distance, which may not be appropriate for all types of data.
- **K-Medoids:** Supports any distance metric, offering greater flexibility.
- **Sensitivity to Outliers**
- **K-Means:** Highly sensitive to outliers, as extreme values can significantly shift the mean.
- **K-Medoids:** More robust to outliers because medoids are real data points and less affected by extreme values.
- **Computational Complexity**
- **K-Means:** Faster and more computationally efficient, making it suitable for very large datasets.
- **K-Medoids:** Slower due to evaluating multiple possible swaps, better suited for smaller datasets or when robustness is important.
- **Convergence**
- **K-Means:** Converges quickly but may get trapped in local minima.
- **K-Medoids:** More likely to reach a global optimum, though at a higher computational cost.

Difference between K means and K medoids Clustering

Aspect	K -Means Clustering	K-Medoids Clustering
Representation of Clusters	K-Means Clustering uses the mean of points (centroid) to represent a cluster.	It uses the most centrally located point (medoid) to represent a cluster.
Sensitivity to Outliers	Highly sensitive to outliers.	More robust to outliers.
Distance Metrics	K-Means primarily uses Euclidean distance.	Whereas it can use any distance metric.
Computational Efficiency	K-Means is generally faster and more efficient	It is slower due to the need to calculate all pairwise distances within clusters.
Cluster Shape Assumption	It assumes spherical clusters.	It does not make strong assumptions about cluster shapes.